

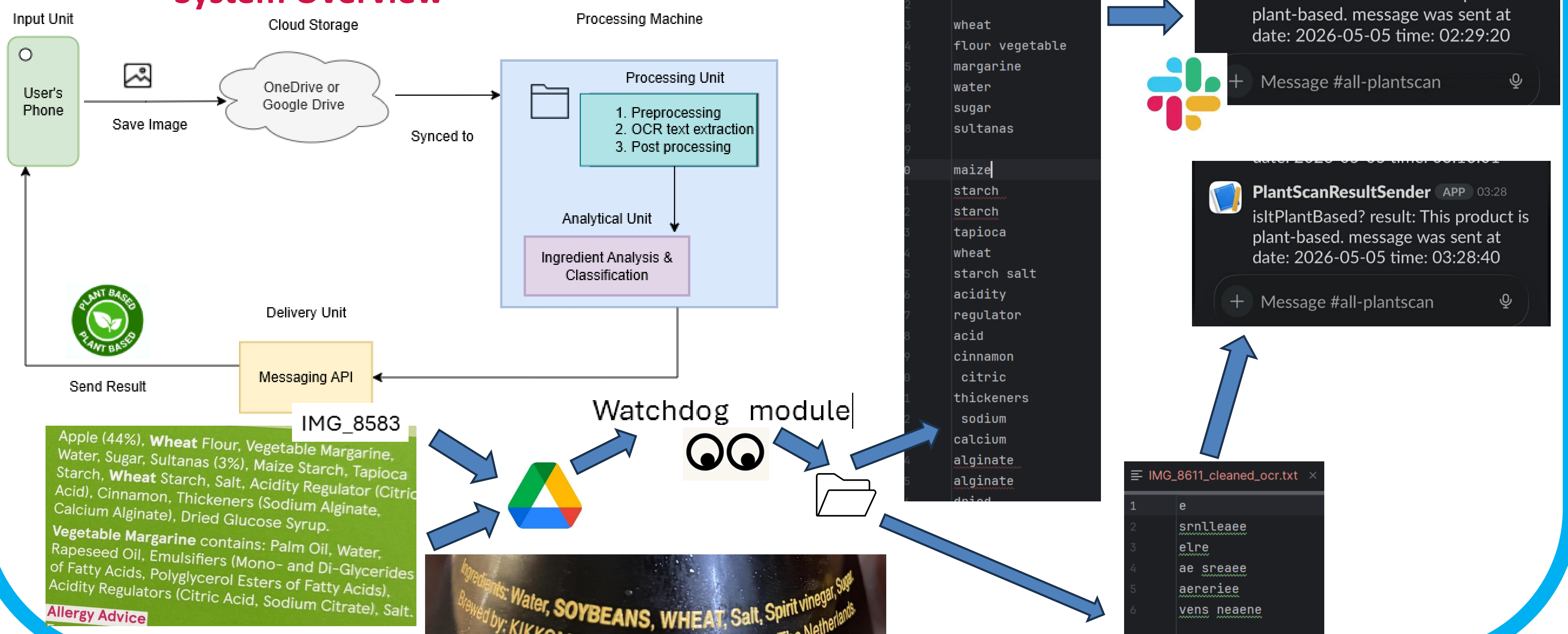
Introduction

This project aims to provide a solution for identifying plant-based food products using Optical Character Recognition (OCR) technology with preprocessing and postprocessing techniques to extract ingredients from food labels, based on a balanced dataset of 60 images. The motivation comes from personal experience. Following a plant-based diet means regularly checking food labels to avoid animal-derived ingredients, a cumbersome process.

Project Goals

- Eliminating the manual effort of reviewing ingredient lists to ensure no animal-derived ingredients are present.
- Implement preprocessing and postprocessing to improve the outcome of text extraction
- Identify products accurately with a rule-based approach

System Overview



OCR Evaluation & Results

Metrics measured at two stages (after preprocessing and after postprocessing).

edit dist = 2 by default	preprocessed	postprocessed	type of metric	ratio	interpretation
cer mean	0.25624746405106763	0.2596084056144458	excellent wer	wer <= 0.02	word error rate is less than or 2%
wer mean	0.4992969307525166	0.4885955739341774	good wer	0.1 >= wer > 0.02	word error rate is between 2-10%
cac mean	0.7437525359489323	0.7403915943855541	poor wer	wer > 0.1	word error rate is more than 10%
wac mean	0.5007030692474834	0.5114044260658226			
#excellent cer	22	20	excellent cer	cer <= 0.02	character error rate is smaller than or 2%
#good cer	6	7	good cer	0.05 > cer > 0.02	character error rate is between 2-5%
#needs review cer	1	2	needs review cer	0.1 >= cer > 0.05	character error rate is between 5-10%
#poor cer	33	33	poor cer	cer > 0.1	character error rate is above 10%
#excellent wer	9	13			
#good wer	10	6	excellent #word edits		1 or 0 word edits need to get gt text
#poor wer	43	43	good #word edits		2-3 words to get gt text
excellent #word edits	13	13	poor #word edits		3+ words to get gt text
good #word edits	9	8			
poor #word edits	40	41			
total_samples	62	62			

2 out of 62 images have 2 versions: a challenging version and a non-challenging of the same image, one from the plant_based group and one from the non_plant_based group.

Conclusions

- Image quality influences OCR outcome (glare on image, curved, wrinkled surfaces, not correctly aligned text, etc.)
- Postprocessing should be improved to apply context aware corrections
- Missing even a single animal-derived ingredient during text extraction can cause a misclassification.

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